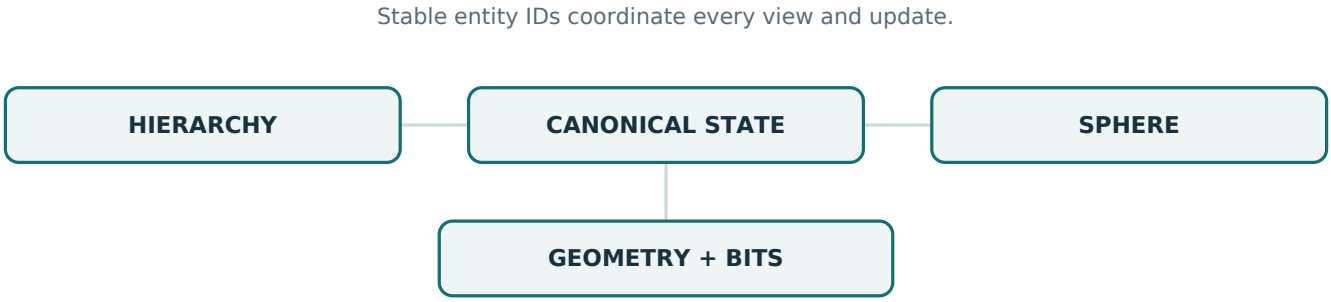
# XD-TAU

## Binary-Spherical Organogram Architecture

One evolving state.
Linked hierarchy, sphere, geometry and memory.

A mathematical specification for navigating populations and organizations through coordinated flat and spherical views.




**WITH THE SHARK-GEYSER AND TAU-AXIS SPIRAL EXTENSION**

**Tom Klootwijk**

Version 1.2 / Reviewed edition / 13 October 2026

Equations, validity conditions and component validation in one specification.

VALIDATION / 02

# Assessment and document guide

**XD-TAU has a coherent mathematical core.** Its binary node numbering, spherical return map, smooth blend and symmetric sign-mask algebra hold under the stated domains. The useful architectural idea is to coordinate these components through stable entity state and explicit mappings.

This edition removes repetitive disclaimer prose and states the working contracts directly. It strengthens spatial-error control, geometric admissibility and boundary ownership, and records what the component checks establish.

| Validation level | Result |
|---|---|
| Algebra and coordinates | Address uniqueness, spherical recovery, mask symmetry and XOR equivalence follow from the definitions. |
| Component checks | 13 checks completed, including counterexample checks. Coverage and results appear on page 14. |
| Executable instance | Select the separator policy, lookup resolution, field definitions, binary schema and update evaluator U. |
| Measured advantage | Evaluate complete navigation and update workloads against established spherical indexes. The benchmark contract is on page 11. |

## Reading order

| Pages | Subject |
|---|---|
| 3-5 | State, partition, binary addresses and organizational layout |
| 6-7 | Spherical coordinates, angular phase and lookup accuracy |
| 8-9 | Signed-distance geometry, blending and admissible fields |
| 10-11 | Packed masks, feedback and application correctness |
| 12-13 | Shark-Geyser stages, spiral relation and view integration |
| 14-15 | Validation record, changes and references |

Equation numbers preserve the v1.1 families where practical. V-numbered expressions are additions made by this review. Definitions specify design choices; identities and counterexamples establish their consequences.

CORE ARCHITECTURE / 03

# The shared state and its views

XD-TAU is a time-stepped information architecture linking a flat hierarchy, spherical regions, a log-polar lookup, geometric fields and encoded binary state.

$$\mathfrak{X} = (\Delta, S_R^2, \delta, \mathcal{L}, \mathcal{G}, \mathcal{B}, \mathcal{U}) \tag{1.1}$$

| Component | Contract |
|---|---|
| Delta | Ordered observations or computational updates. |
| Sphere $S_R^2$ | Spatial domain for entity positions and navigation. |
| Binary hierarchy delta | Partition nodes, parent/child relationships and logical addresses. |
| Lookup L | Explicit mappings among nodes, entities, display positions and spherical regions. |
| Geometry G | Primitives, blend operators and Klein-derived solid geometry. |
| Binary representation B | Encoded data, parameters and operator descriptions. |
| Evaluator U | A selected update rule that produces the next valid state. |

## Identity coordinates the representations

$$e = (\mathrm{ID}, \mathrm{orgParent}, \mathrm{spatialAddress}, \mathrm{position}, \mathrm{payload}) \tag{V.1}$$

The entity ID is stable. A spatial address identifies a cell, and multiple entities can occupy that cell. Organizational parents are stored separately from spatial partitions. An organization can have any number of direct reports; a binary display or index retains that original relation.

Selecting an organizational subtree resolves a set of entity IDs, which can occupy many spherical cells. A position change updates spatial information. A reporting-line change updates the organizational relation. Each operation preserves the other information unless the event explicitly changes it.

> **Canonical state**
>
> Every view resolves through shared IDs and declared mappings. The canonical record carries the information needed to preserve identity, payload and spatial accuracy through updates.

CORE ARCHITECTURE / 04

# Time, hemispheres and refinement

Use separate symbols for a full turn, an update step and the pinion phase:

$$\tau_\circ = O = 2\pi, \qquad \mathcal{T}_n = \text{step } n, \qquad \omega_n = \text{pinion phase} \tag{2.1}$$

$$\Delta : n \longrightarrow n+1, \qquad T_n = \ln(1 + t_n/t_*), \quad t_n \geq 0,\ t_* > 0 \tag{2.2}$$

$T_n$ is a dimensionless log-time coordinate. Chronological use requires the sequence $t_n$ to be nondecreasing; the update index n supplies the computational order.

## The first cut

$$S_R^2 = \{(x,y,z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = R^2\}, \quad R > 0 \tag{2.3}$$

$$\Pi_0 = S_R^2 \cap \{y < 0\}, \qquad \Pi_1 = S_R^2 \cap \{y \geq 0\} \tag{2.4}$$

The first address bit is 0 for y < 0 and 1 for y ≥ 0. This gives disjoint ownership of the hemispherical boundary. The apex A = (0,0,R), an equatorial circular base and a cone bisected by y = 0 retain the source geometry; the cone has a triangular axial section.

## Recursive separators

$$C_\varnothing = S_R^2, \qquad g_w(p) = \mathbf{n}_w \cdot p - c_w, \quad \|\mathbf{n}_w\| = 1 \tag{2.5a}$$

$$C_{w0} = C_w \cap \{g_w < 0\}, \qquad C_{w1} = C_w \cap \{g_w \geq 0\} \tag{2.5b}$$

Active leaf cells cover the sphere with unique boundary ownership. A separator policy selects nonempty children and its balancing measure: spherical area, population count or workload. Equal-area splitting uses $\mu(C_{w0}) = \mu(C_{w1}) = \mu(C_w)/2$.

> **Spatial refinement contract**
>
> Each cell representative must meet the declared maximum surface-distance error. Equal area alone permits long, thin cells; spatial precision therefore uses cell radius or retained entity coordinates. Time step and tree depth remain separate indices.

CORE ARCHITECTURE / 05

# Binary addresses and the flat layout

A node at depth d has a finite binary word w = $b_1 \ldots b_d$. Removing its final bit gives its parent; appending 0 or 1 gives its children. A minimal L-system production is:

$$N(w) \longrightarrow N(w0)N(w1) \tag{3.1}$$

$$j(w) = \sum_{i=1}^{d} b_i 2^{d-i}, \qquad \mathrm{addr}(w) = 2^d - 1 + j(w) \tag{3.2, 3.5}$$

$$2^d - 1 \le \mathrm{addr}(w) \le 2^{d+1} - 2 \tag{V.2}$$

Depth ranges are disjoint, so every finite word has a unique logical slot. If a node has slot a, its children have slots 2a+1 and 2a+2. Sparse allocation uses a node table keyed by this logical slot.

## Log-polar organogram

$$I_w = \left[\frac{2\pi j(w)}{2^d}, \frac{2\pi(j(w)+1)}{2^d}\right) \tag{3.2}$$

$$\theta_w = \frac{2\pi(j(w)+1/2)}{2^d}, \qquad r_d = r_0 a^d, \quad r_0 > 0,\ a > 1 \tag{3.3}$$

$$\rho_d = \ln(r_d/r_0) = d \ln a, \qquad \mathcal{L}_{\mathrm{org}}(w) = (\rho_d, \theta_w) \tag{3.4}$$

Hierarchy levels are equally spaced in log radius; branching halves angular sectors. Positions, segment lengths and appearance belong to the geometric interpretation of the production. The organogram radius expresses depth, while the spherical chart radius expresses projected position.

## Organization and search semantics

The binary spatial hierarchy and the organizational parent graph are linked by entity IDs. A binary layout of a nonbinary organization can insert display grouping nodes while retaining the actual organizational parents in the records.

An ordered search-tree implementation also declares a key and its subtree ordering rule. Spatial branching supplies a partition hierarchy; search ordering is an additional data-structure contract.

> **Validated address result**
>
> Exhaustive enumeration through depth 16 checked 131,071 distinct slots and both child formulas without a collision.

CORE ARCHITECTURE / 06

# Spherical coordinates and OTAN2

Let h ∈ {0,1} identify the y hemisphere, $s_h = 2h-1$, and use angular coordinates φ and ψ:

$$p = R(\sin\varphi\cos\psi,\ s_h\cos\varphi,\ \sin\varphi\sin\psi) \tag{4.1}$$

$$0 \le \varphi \le \pi/2, \qquad 0 \le \psi < 2\pi \tag{4.2}$$

At φ = π/2, only h = 1 owns the seam. At φ = 0, a designated angular code represents the pole.

$$r = \sqrt{x^2+z^2} = R\sin\varphi, \qquad \rho = \ln(1 + r/r_*), \quad r_* > 0 \tag{4.3-4.4}$$

$$0 \le \rho \le \ln(1 + R/r_*) \tag{4.5}$$

## Turn-valued phase

$$q = \mathrm{OTAN2}(z, x; T_n) = \mathrm{frac}\left(\frac{\mathrm{atan2}(z,x) + \omega_n}{2\pi}\right) \tag{4.6}$$

$$\omega_n = [\omega_0 + \nu T_n]_{2\pi} \tag{4.7}$$

q is measured in turns. The phase advances at rate ν per unit of log-time. Encode and decode use the same phase or phase epoch; a changed display phase refreshes the code while preserving the entity position.

$$\mathrm{atan2}(cz, cx) = \mathrm{atan2}(z, x), \qquad c > 0 \tag{4.8}$$

Common positive scaling preserves the angle, so the explicit phase supplies its time dependence.

## Return map

$$r = r_*(e^{\rho} - 1), \qquad \psi = [2\pi q - \omega_n]_{2\pi} \tag{4.9}$$

$$p = (r\cos\psi,\ s_h\sqrt{R^2 - r^2},\ r\sin\psi), \quad 0 \le r \le R \tag{4.10}$$

$$\|p\|^2 = r^2\cos^2\psi + (R^2 - r^2) + r^2\sin^2\psi = R^2 \tag{4.11}$$

These equations recover valid coordinate representatives. The pinion controls angular addressing or view rotation; movement events update recorded spatial coordinates.

CORE ARCHITECTURE / 07

# Lookup accuracy and boundary contracts

A lookup entry identifies the cell word, hemisphere, coordinate convention, phase epoch, entity IDs and position precision. Region representatives support aggregates. Entity coordinates or residuals preserve more detailed positions.

## Surface metric

$$d_S(p,q) = R\,\mathrm{atan2}(\|p \times q\|, p \cdot q), \qquad p,q \in S_R^2 \tag{V.3}$$

$$\sup_{p \in C_w} d_S(p, \widehat{p}_w) \le \varepsilon_{\mathrm{space}} \tag{V.4}$$

A representative-based cell is acceptable when (V.4) holds. A point record can instead meet the bound through its own stored coordinates. For a sequence of spatial refinements, require the chosen error bounds to decrease to the intended resolution.

## Why area balance needs this extra condition

The area of a spherical latitude belt of y-width Δy is 2πRΔy. Repeated y-directed cuts can reduce that area to zero while a belt adjacent to the equator still has geodesic diameter πR. Antipodal equator points can retain the same address through every refinement. The error criterion prevents that failure.

| Boundary or precision case | Implementation rule |
|---|---|
| Pole r = 0 | Use one designated angular code; derive or retain the hemisphere. |
| Hemisphere seam r = R | Use h = 1 ownership and explicit cross-seam neighbors. |
| Cell separator $g_w = 0$ | Use the right-child ownership convention consistently. |
| Floating-point roundoff | Accept only a specified numerical tolerance; bound any correction to the legal radius interval. |
| Near the seam | Retain residual information or use a chart with better conditioning when required by the error budget. |

The inverse radial derivative $d\varphi/dr = 1/\sqrt{(R^2 - r^2)}$ grows near the seam. In double precision, a point with R = 1 and $y = 10^{-9}$ can have r rounded to 1, causing reconstructed y = 0. This is a numerical conditioning issue, handled by the precision contract.

**Component result**

For 100,000 random unit-sphere points at a shared phase, the maximum measured round-trip surface error was $8.29 \times 10^{-11}$ radians. Six exact pole/seam cases also passed. These checks measure coordinate recovery; they do not bound every quantized LUT.

CORE ARCHITECTURE / 08

# Distance fields and the blend operators

Geometry construction uses scalar fields with compatible length units. Occupancy defines a region; the signed-distance layer measures the nearest region boundary, with negative values inside.

$$d_\Omega(p) = -\mathrm{dist}(p, \partial\Omega) \quad (p \in \Omega) \tag{5.1a}$$

$$d_\Omega(p) = +\mathrm{dist}(p, \partial\Omega) \quad (p \notin \Omega) \tag{5.1b}$$

Admissible geometry uses a closed region Ω with a nonempty boundary. This makes $\Omega = \{p : d_\Omega(p) \le 0\}$, including its boundary.

## Hard and encircled blends

$$a \boxplus b = \min(a, b) \tag{5.3}$$

Keep the 45-degree angle $\alpha = \pi/4$ distinct from the smoothing length $h_b > 0$. Rotate the scalar field-value coordinates:

$$u = (a + b)/\sqrt{2}, \qquad v = (b - a)/\sqrt{2} \tag{5.4a}$$

$$a \oplus_{h_b} b = \frac{u - \sqrt{v^2 + h_b^2}}{\sqrt{2}} \tag{5.4b}$$

$$S_{h_b}(a, b) = \frac{a + b - \sqrt{(a - b)^2 + 2h_b^2}}{2} \tag{V.5}$$

$$0 < \min(a, b) - S_{h_b}(a, b) \le h_b/\sqrt{2} \tag{V.6}$$

The encircled blend is a smooth minimum converging to min(a,b) as $h_b$ approaches zero. Its maximum downward displacement is $h_b/\sqrt{2}$, attained at a = b. The field-value rotation specifies the algebra; a physical cutting plane is supplied by a geometric separator.

## Region first, distance second

Blends and modulation construct the implicit region. Reconstructing $d_\Omega$ afterward gives the actual boundary distance. For example, with Φ(x,y,z) = x, the expression $\mathrm{sgn}(\Phi)\|\nabla\Phi\|$ returns sgn(x), while the actual signed distance is x.

Euclidean distance in three-dimensional geometry and geodesic distance on the navigation sphere use their respective metrics.

CORE ARCHITECTURE / 09

# Accumulated geometry and valid states

Choose three-dimensional primitive fields $f_{pyr}$, $f_{cone}$ and $f_{sphere}$. A triangular side profile receives an explicit solid realization, such as an extrusion. Specify each primitive's dimensions, orientation, units and evaluation method.

## Klein-derived solid

$$\iota : K \longrightarrow \mathbb{R}^3, \qquad \Omega_{K,\varepsilon} = \{p : \operatorname{dist}(p, \iota(K)) \le \varepsilon\}, \quad \varepsilon > 0 \tag{5.6}$$

$$f_K = d_{\Omega_{K,\varepsilon}} \tag{5.7}$$

A selected immersion, scale and thickness define a closed thickened image and its distance. The abstract Klein surface supplies the source geometry; its immersed image supplies the region used here.

## Accumulation

$$m_n(p) = M_{i_n(p),\, j_n(p)} \in \{-1,\ +1\} \tag{5.8}$$

$$\Phi_n = (f_{\mathrm{pyr}} \boxplus f_{\mathrm{cone}}) \oplus_{h_b} (f_{\mathrm{sphere}} \boxplus [m_n f_K]) \tag{5.9}$$

$$\Omega_n = \overline{\{p : \Phi_n(p) \le 0\}}, \qquad d_n = d_{\Omega_n} \tag{V.7}$$

All fields are evaluated at p. The closure in (V.7) makes boundary occupancy explicit when a lookup mask is discontinuous. It is an edition 1.2 refinement of the region rule in v1.1.

> **Geometric admissibility invariant**
>
> Before sampling or committing an update, establish that the region is closed, its boundary is nonempty, and the selected distance evaluator is defined and meets its error budget. Invalid proposals retain the last valid state and return a diagnostic.

## Why the sign mask needs this invariant

Multiplying $f_K$ by $-1$ reverses its inside/outside sign. If the sphere solid contains $\Omega_{K,\varepsilon}$ and a permitted lookup selects $m = -1$ everywhere, then $\min(f_{sphere}, -f_K) \le 0$ at every point: the sphere covers its interior, and $-f_K$ covers its exterior. The outer smooth minimum makes $\Omega_n$ all of space. Its empty boundary is outside the distance layer's domain.

A concrete implementation can restrict the mask, clip to a declared bounded domain with an explicit boundary, or reject this state through the invariant. The chosen rule forms part of the geometry configuration.

CORE ARCHITECTURE / 10

# Double Hadamard products and packing

For square sign matrices A,B ∈ $\{-1,+1\}^{N\times N}$, retain the double elementwise product:

$$M = (A \odot B) \odot (A^T \odot B^T) \tag{6.1}$$

$$C = A \odot B, \qquad M = C \odot C^T, \quad M_{ij} = C_{ij}C_{ji} \tag{6.2a}$$

$$M_{ij} = M_{ji}, \qquad M_{ii} = 1 \tag{6.2b}$$

This constructs a symmetric sign mask with a unit diagonal. It can represent undirected pairwise modulation. If A = B, all entries of C and M are +1.

## Bitwise form and information content

$$s = (-1)^b, \quad b \in \{0,1\}, \qquad (-1)^a(-1)^b = (-1)^{a \,\mathrm{XOR}\, b} \tag{6.3}$$

$$m_{ij}^{\mathrm{bit}} = a_{ij} \,\mathrm{XOR}\, b_{ij} \,\mathrm{XOR}\, a_{ji} \,\mathrm{XOR}\, b_{ji} \tag{6.4}$$

$$K_{\mathrm{mask}} = \frac{N(N-1)}{2}, \qquad \#\{\text{possible } M\} = 2^{K_{\mathrm{mask}}} \tag{V.8}$$

Every upper-triangular off-diagonal bit is independently selectable; the lower triangle and diagonal are determined. Each mask has $2^{(3N^2+N)/2}$ input preimages among the pairs (A,B). The mask therefore retains pairwise sign information while discarding other input distinctions.

## Packed storage

$$\mathrm{bit}(W,i) = (W_{\lfloor i/64 \rfloor} \gg (i \bmod 64)) \,\mathrm{AND}\, 1 \tag{6.5}$$

Pack the two input bit planes separately when both inputs must remain recoverable. Their allocated size is $128\lceil N^2/64 \rceil$ bits. A mask-only representation stores the $K_{mask}$ upper-triangle bits in $64\lceil K_{mask}/64 \rceil$ bits. Records also declare lengths, offsets, bounds and a versioned decoding schema.

**Component result**

All 512 possible 3×3 C matrices produced exactly eight masks, matching $K_{mask} = 3$. An additional 4,096 random 16×16 input pairs satisfied the XOR identity. A 257-bit record passed retrieval across 64-bit word boundaries.

CORE ARCHITECTURE / 11

# Feedback, correctness and application

Let $B_n$ be the finite encoded state and $e_n$ an external event. Decoded state includes entity records, the partition, lookup configuration, primitive and phase parameters, mask inputs and an update description.

$$s_n = \mathrm{Decode}(B_n), \qquad p_{n,i} = \mathcal{L}(s_n, i) \tag{7.1a}$$

$$q_{n,i} = \mathbf{1}[d_{s_n}(p_{n,i}) + J_{n,i} \le 0], \quad |J_{n,i}| \le \varepsilon_J \tag{7.1b}$$

$$s_{n+1} = \mathcal{U}(s_n, e_n, q_n), \qquad B_{n+1} = \mathrm{Encode}(s_{n+1}) \tag{7.1c}$$

U evaluates the current encoded rule description, uses the samples and event, validates the proposed state, then commits it atomically. Semantic encoding satisfies Decode(Encode(s)) = s. IDs and unrelated payloads survive every committed update. Reproducible sampling retains a fixed jitter rule and generator state.

The sample bit classifies interior or exterior. Locations, parameters and numerical records carry the geometry detail. Whenever $|d(p)| > \varepsilon_J$, every permitted jitter gives the same classification.

## Feedback behavior

For a fixed update rule, fixed inputs and bounded finite state, deterministic iteration eventually becomes periodic. Convergence to a fixed point is established separately. One sufficient proof is a nonnegative integer rank V that strictly decreases on every nonstationary semantic update under frozen inputs. A one-bit flip rule gives a two-cycle.

## Navigation and acceptance

Select an entity or organizational subtree; resolve its IDs and spatial addresses; retrieve its spherical representation; navigate or apply an event; commit validated state; refresh the linked views. Acceptance checks cover identity preservation, unique cell ownership, cross-seam neighbors, spatial-error bounds and deterministic replay.

## Performance evaluation

Measure build cost, memory, median and p95 selection latency, movement updates and organizational subtree operations at equal spatial accuracy. Use uniform, clustered, skewed and seam-heavy data. Include S2 or HEALPix-backed shared-ID implementations as baselines [R1-R2]. Compare runs with and without field evaluation, mask generation and feedback to identify each component’s contribution.

A depth-d traversal takes d branch decisions. Logarithmic item-count depth depends on the balancing rule. Full performance results include lookup construction, storage, updates and distance reconstruction.

OPTIONAL EXTENSION / A / 12

# Shark-Geyser and radial encoding

The extension supplies a derived geometric display in four stages: triangular folding, chamber/eruption, barycentric log-polar encoding and Len-Theta output. Its settings declare the missing coupling functions before a particular renderer or simulation is evaluated.

## 1. Shark profile

$$f_p(x) = |(x \bmod p) - p/2|, \qquad p > 0 \tag{A.1}$$

This triangular periodic wave has range [0,p/2]. It discards the period index and branch side. A reversible use retains those values as metadata; a visual modulation can use the folded value directly.

## 2. Chamber and eruption

The chamber stage supplies a selected accumulation function. The eruption stage is:

$$r' = r_0 \exp(\alpha_g (r - r_c)), \qquad r_0 > 0 \tag{A.2}$$

$$r = r_c + \frac{\ln(r'/r_0)}{\alpha_g}, \qquad \alpha_g \neq 0 \tag{V.9}$$

With length-valued radius, $\alpha_g$ has inverse-length units. Positive $\alpha_g$ gives increasing expansion; zero $\alpha_g$ gives a constant output. The renderer declares its radius bounds. Applying a logarithm directly afterward gives $\ln(r'/r_0) = \alpha_g(r - r_c)$, an affine radial coordinate.

## 3. Barycentric log-polar coordinates

$$\lambda_1 + \lambda_2 + \lambda_3 = 1, \qquad u = \lambda_1 v_1 + \lambda_2 v_2 + \lambda_3 v_3 \tag{V.10}$$

$$\rho_{\text{raw}} = \ln(r/r_*), \qquad r > 0,\ r_* > 0 \tag{A.3*}$$

Specify a nondegenerate reference triangle $(v_1, v_2, v_3)$. Nonnegative weights give triangle containment. The reference radius $r_*$ makes the logarithm dimensionless. A complete encoding declares the map from the angular/radial domain to u and the resulting weights.

## 4. Len-Theta output

Declare Len as a dimensionless output of the radial and folding stages and Theta as the accumulated angular phase. Their coupling, range and inverse metadata are module parameters. Core spatial identity and coordinates remain available alongside this output.

OPTIONAL EXTENSION / A / 13

# Tau-axis spiral and view integration

Select a real eigenvector $v_{tau} = (x_{tau}, y_{tau})$ of a specified matrix, with an eigenvalue, normalization and sign rule. Its direction is $\beta = \operatorname{atan2}(y_{tau}, x_{tau})$.

$$\Theta(\theta) = \theta + \alpha_\theta \tanh\left(\frac{\theta - \beta}{\sigma}\right) \sin(2\pi\,\mathrm{Len}), \quad \sigma > 0 \tag{A.4}$$

For fixed Len, put $a = \alpha_\theta \sin(2\pi \mathrm{Len})$. Then $\Theta' = 1 + (a/\sigma)\operatorname{sech}^2((\theta-\beta)/\sigma)$. The sufficient local monotonicity condition $|\alpha_\theta| < \sigma$ applies for fixed Len. A circular display also declares its seam rule; the tanh perturbation is generally nonperiodic.

At $\alpha_\theta = 2$, $\sigma = 1$, Len = 3/4 and $\beta = \pi$, the three angles 1.2265846054, π and 5.0566007017 all map to π. Retained IDs and overlap-aware picking handle such folds.

## Spiral relation and scale repetition

$$z = k\Theta, \qquad I(r) = \sin(\omega_s \ln(r/r_*)) \tag{A.5*}$$

$$I(qr) = I(r), \qquad q = \exp(2\pi/|\omega_s|), \quad \omega_s \neq 0 \tag{V.11}$$

The screw uses unwrapped phase or a stored turn count; k has units of axial length per radian. The log-sine has exact discrete scale repetition. A finite renderer specifies $0 < r_{min} \le r \le r_{max}$ and samples enough cycles to control aliasing. Define how negative I values affect radius, pitch or visibility.

## Attenuation and motion

$$A(t) = A_0 e^{-\mu t}, \quad \mu \ge 0, \qquad v \cdot (\pm 2\Omega \times v) = 0 \tag{A.6, A.8}$$

The exponential defines the attenuation of the named quantity A. A spatial diffusion model supplies its own evolution equation. A Coriolis-style deflection is perpendicular to velocity and performs zero work through that term. The material analogy can guide visual styling.

## Module boundary

$$\widetilde{V}_n = \mathcal{G}_\eta(\mathrm{View}(B_n)), \quad \mathcal{G}_{\mathrm{off}} = \mathrm{id}, \quad B_{n+1} = \mathcal{F}(B_n, e_n) \tag{A.7}$$

The extension derives a view and retains original IDs and coordinates. Its output stays outside the core feedback path. Picking returns retained identities and resolves overlap through candidate lists or layer order. Reproducing the view stores both $B_n$ and the module settings η.

REVIEW / 13 OCTOBER 2026 / 14

# Validation record

The review combines algebraic derivations with seeded numerical checks. The 13 checks below all returned their expected result; counterexample checks confirm specific failure cases rather than general system success.

| Check | Coverage | Observed result |
|---|---|---|
| Node addresses | All nodes, depths 0-16 | 131,071 unique addresses; child formulas hold. |
| Sphere round trip | 100,000 random unit points | Max surface error $8.29\times10^{-11}$ rad. |
| Seam conditioning | R=1; y=$10^{-9}$ | Projected-radius rounding can lose y. |
| Poles and seam | Six exact cases | Designated codes recover each point. |
| Mask capacity | All 512 C matrices, N=3 | Eight masks; exactly three free bits. |
| Mask XOR form | 4,096 pairs, N=16 | Matches sign multiplication. |
| Packed retrieval | 257 bits in five words | All bits recovered across word boundaries. |
| Smooth-min bound | 100,000 random pairs | Max displacement/h=0.706967; bound=0.707107. |
| Angular fold / seam | Explicit fixed-Len example | Three-to-one collision; seam advance 2.298097 rad. |
| Log-sine repetition | 10,000 radii; omega=3.25 | Max residual $3.50\times10^{-15}$ at ratio 6.912199. |
| Geyser inverse | 1,000 radii | Max reconstruction error $4.44\times10^{-16}$. |
| Area / resolution | Equal-area belt argument | Area→0 with diameter πR. |
| Feedback cycle | One-bit flip update | Orbit 0,1,0,1,0. |

Numerical environment: Python, NumPy and IEEE 754 double precision; seed 20261013. Random sampling verifies these instances. The address, mask and blend identities and the displayed counterexamples have separate algebraic arguments.

**Assessment**

The formal core supports linked hierarchy/sphere navigation, exact node addressing, explicit coordinate recovery, geometric evaluation and encoded feedback. A runnable instance is completed by selecting its algorithms and enforcing the spatial and geometric validity contracts. System performance and convergence are evaluated for that selected instance.

PROVENANCE / 15

# Edition record and references

## What changed in version 1.2

The core vocabulary, architecture and equation families are retained. Repetitive “proposed,” “not automatically,” and source-status caveats are replaced by direct definitions and implementation contracts. Validation results are consolidated on page 14.

The revision adds a geodesic cell-error requirement, shared phase/epoch handling, closed-region occupancy and a nonempty-boundary invariant. It simplifies the smooth blend, states its displacement bound, quantifies mask information content, and records addendum folding and discrete scale repetition. The radial addendum logarithms now include explicit reference scales.

The region closure in (V.7) and validation of state proposals are substantive specification refinements. The separator policy, primitive realization, lookup resolution, binary schema, chamber function, Len mapping and evaluator U remain implementation parameters to be instantiated.

## Document lineage

Primary reviewed source: *Tom Klootwijk, XD-TAU: Binary-Spherical Organogram Architecture*, version 1.1, 13 October 2026, 16 pages. This edition assesses its displayed definitions and equations.

Version 1.1 identifies three prior conversation-export sources: S1, organic organogram and log-polar architecture (14 pages); S2, XD-TAU and the first hemispherical cut (9 pages); and S3, *geiser-haailen-samen.pdf* (6 pages), plus F0, a preceding formalization. This edition follows their description in v1.1; its direct validation uses the equations printed in v1.1.

## Primary technical references

[R1] S2 Geometry, S2 Cells. Hierarchical spherical cells, unique identifiers and coordinate mappings. s2geometry.io/devguide/s2cell_hierarchy

[R2] HEALPix, Geometric and Algebraic Properties. Equal-area spherical pixels and hierarchical subdivision/numbering. healpix.sourceforge.io - geometric and algebraic properties

These references establish relevant spherical-indexing baselines. XD-TAU’s distinctive design contribution is its stated integration and interaction workflow; claims of originality or superiority require the corresponding comparison.

## Attribution

Paradigm attribution: Tom Klootwijk. Mathematical review, editorial revision and component-check code prepared with AI assistance. This edition is a reviewed specification; the original v1.1 file remains intact.